

Relational Schema

Entities:

User(email, phone, name, date_of_birth)

Student(email, payment_info, edu_level)
email refers to user

Tutor(email, bank_info)
email refers to user

Post(pID, title, post_date, body, email)
email refers to user

post_comment(cID, email, pID, comment_date, text)
email refers to user
pID refers to post

Weak entities:

Service(email, name, subject, price, edu_level)
email refers to Tutor

Notes_set(email, name, course, document)
email, name refers to service

Tutorial_session(email, name, tutorial_date, tutorial_time, location, max_participants)
email, name refers to service

Relationships:

Purchase(email_buyer, email_seller, name, purchase_date)
email_buyer refers to Student
email_seller, name refers to service

Advertises(pID, email, name)
pID refers to post
email refers to tutor
email, name refers to service

post_like(email, pID)
Email refers to user
pID refers to post

comment_like(email, cID)
Email refers to user
cID refers to post_comment

Pending Constraints

- User advertisement posts can only be done for services that they have purchased
- Once max participants of tutorial session is reached, it becomes full (unpurchasable)
- Bank_info and payment_info should be valid bank information.

Application Procedure

Our simple menu consists of printing to the console and contains 5 database options as well as an additional option to quit the program.

```
Connected to the database.

Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: |
```

Option 1: View a Tutors services.

When selected this option prompts the user to input the desired tutors email (which is their primary key in the database) and then displays the names of all the services that the tutor offers.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 1
You selected option 1
Input a Tutor email:
ben@mail.com
name: Basic Algebra
name: Calculus Cram Course
name: Economic Theory
name: Free Tryout for Trig
name: Full Course on Trig Functions
name: Functions notes p1
name: Functions notes p2
name: Functions notes p3
name: Wednesday Math Tutorials
```

Option 2: Search for services by subject and education level

This option when selected will prompt a user for a subject and then an education level. Once those values are inputted it will list all services offered by any tutor with the matching subject and education level.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 2
You selected option 2
What subject are you studying?
Trigonometry
What is the desired education level?
High School
Tutor: david14@mail.com Title: Monday Math Tutorials Price: 25$
Tutor: david14@mail.com Title: Trig Notes 1/2 Price: 7$
Tutor: david14@mail.com Title: Trig Notes 2/2 Price: 7$
Tutor: ben@mail.com Title: Free Tryout for Trig Price: 0$
Tutor: ben@mail.com Title: Full Course on Trig Functions Price: 33$
```

Also, if you look for services but there are none offered for that subject and education level the console will output the following message:

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 2
You selected option 2
What subject are you studying?
Archaic Languages
What is the desired education level?
University
Sorry there are no services for that subject and education level!
```

Option 3: Delete a user post.

When selected this option will first prompt for the email associated to the user whos post we would like to delete. Once the email is entered a secondary menu opens up with a list of the posts by that user. Then one must select the index of the post they wish to delete from the second menu.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 3
You selected option 3
Input the email of the user who made the post:
ben@mail.com
Below are the posts made by that user, please select the one you wish to delete
1. pid: 2 title:Should I make a course for Vectors
2. pid: 3 title:Should I make a course for Matrices
3. pid: 4 title:Should I make a course for Algebra
4. pid: 6 title:Functions
5. pid: 12 title:Free Tryout for Trig
6. pid: 13 title:Full Course on Trig Functions
7. pid: 14 title:post title 14
8. pid: 17 title:post title 17
9. pid: 24 title:Reading Comprehension for Elementary
10. pid: 30 title:Study Skills for High School
11. pid: 35 title:Middle School Algebra Tricks
12. pid: 41 title:Undergraduate Economics
13. pid: 43 title:Why Algebra Matters in Middle School
14. pid: 46 title:Economic Theory Workshop this Weekend
15. pid: 52 title:Calculus Cram Course: Now Available Online
To select the post please input the index of the post displayed on screen, to abort enter -1
1
```

As we can see below if we run the same command again the post with pid=2 does not appear in the list of posts we can delete, indicating it has been deleted from the database as intended.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 3
You selected option 3
Input the email of the user who made the post:
ben@mail.com
Below are the posts made by that user, please select the one you wish to delete
1. pid: 3 title:Should I make a course for Matrices
2. pid: 4 title:Should I make a course for Algebra
3. pid: 6 title:Functions
4. pid: 12 title:Free Tryout for Trig
5. pid: 13 title:Full Course on Trig Functions
6. pid: 14 title:post title 14
7. pid: 17 title:post title 17
8. pid: 24 title:Reading Comprehension for Elementary
9. pid: 30 title:Study Skills for High School
10. pid: 35 title:Middle School Algebra Tricks
11. pid: 41 title:Undergraduate Economics
12. pid: 43 title:Why Algebra Matters in Middle School
13. pid: 46 title:Economic Theory Workshop this Weekend
14. pid: 52 title:Calculus Cram Course: Now Available Online
To select the post please input the index of the post displayed on screen, to abort enter -1
-1
```

To ensure proper execution, if the user inputs a bad option we get the following output:

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 3
You selected option 3
Input the email of the user who made the post:
ben@mail.com
Below are the posts made by that user, please select the one you wish to delete
1. pid: 3 title:Should I make a course for Matrices
2. pid: 4 title:Should I make a course for Algebra
3. pid: 6 title:Functions
4. pid: 12 title:Free Tryout for Trig
5. pid: 13 title:Full Course on Trig Functions
6. pid: 14 title:post title 14
7. pid: 17 title:post title 17
8. pid: 24 title:Reading Comprehension for Elementary
9. pid: 30 title:Study Skills for High School
10. pid: 35 title:Middle School Algebra Tricks
11. pid: 41 title:Undergraduate Economics
12. pid: 43 title:Why Algebra Matters in Middle School
13. pid: 46 title:Economic Theory Workshop this Weekend
14. pid: 52 title:Calculus Cram Course: Now Available Online
To select the post please input the index of the post displayed on screen, to abort enter -1
20
Invalid choice please try again
-1
```

Furthermore, if there are no posts associated to the input email we get the following message:

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 3
You selected option 3
Input the email of the user who made the post:
fake@mail.com
That user has no posts sorry!
```

Option 4: Purchase a service

This option when selected prompts the user for the email and name of the service (which combined make the primary key of the service in the database). Then there is a check done to make sure that either the service being searched is a notes set in which case it can always be

purchased, or if it is a tutorial that the tutorial still has open spots left in it. If there were no open spots in the tutorial we would get the following output:

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 4
You selected option 4
Input the email of the Tutor who offers the service:
max@mail.com
Input the name of the service:
Tuesday Chem Tutorials
The number of people who have purchased this service: 3
Unfortunately this service is full
```

When trying again with a tutorial that does have open spots we are then prompted to input our email to sign ourselves up for the service.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 4
You selected option 4
Input the email of the Tutor who offers the service:
max@mail.com
Input the name of the service:
Tuesday Chem Tutorials2
The number of people who have purchased this service: 0
The service is available for purchase, Enter y to continue, n to abort
y
Please input your user email
david10@mail.com
```

And now when we check the service again we can see that the number of people who have purchased the service went up by 1 indicating the purchase was added in the database.


```

Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 4
You selected option 4
Input the email of the Tutor who offers the service:
max@mail.com
Input the name of the service:
Tuesday Chem Tutorials2
The number of people who have purchased this service: 1
The service is available for purchase, Enter y to continue, n to abort

```

You can also abort a purchase without issue

```

Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 4
You selected option 4
Input the email of the Tutor who offers the service:
max@mail.com
Input the name of the service:
Tuesday Chem Tutorials2
The number of people who have purchased this service: 1
The service is available for purchase, Enter y to continue, n to abort
n
aborting purchase

```

Option 5: View who liked a post

The option simply queries the user for the pID of the post they wish to view the likes of and then outputs the emails of all the users who liked that post.

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 5
You selected option 5
Please input the pID of the post:
1
david10@mail.com
david11@mail.com
david12@mail.com
david1@mail.com
david2@mail.com
david3@mail.com
david4@mail.com
david5@mail.com
david6@mail.com
david7@mail.com
david8@mail.com
david9@mail.com
```

And if the user inputs an invalid pID or if the post has no likes we get the following output:

```
Menu:
1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit
Select an option: 5
You selected option 5
Please input the pID of the post:
1234
Sorry, we cannot find any likes on a post with that ID
```

Option 6: Quit

This option just stops the execution of the java program

Menu:

1. View a Tutors services
2. Search for services by subject and education level
3. Delete a user post
4. Purchase a service
5. View who liked a post
6. Quit

Select an option: 6

Exiting program.

Process finished with exit code 0

Stored procedure

We created a stored procedure that is intended to be used when a student uses the search function to find services. The procedure takes as input the subject that the student searched for, then creates automatically generated posts that recommend them some tutors that are well known within the community. The criteria we chose for this was having at least one post with a score greater than five when we add the number of likes to twice the number of comments.

The code was too long to include in this document, so please consult the file `stored_procedure.sql` included in the repository.

Query of the post table before running the procedure:

Running the procedure, then querying the post table:

```
db2 => select * from post p where p.post_date in (select current date from sysibm.sysdummy1)
```

PID	TITLE	POST_DATE	BODY	EMAIL
53	Trigonometry: From Basics to Advanced	03/25/2025	Comprehensive trigonometry course for high school students	david14@mail.com

```
1 record(s) selected.
db2 => call make_advert('Trigonometry')
Return Status = 0
db2 => select * from post p where p.post_date in (select current date from sysibm.sysdummy1)
```

PID	TITLE	POST_DATE	BODY	EMAIL
53	Trigonometry: From Basics to Advanced	03/25/2025	Comprehensive trigonometry course for high school students	david14@mail.com
55	Today's recommended tutor: david14@mail.com	03/25/2025	Looking for help with Trigonometry? david14 has you covered!	ADMIN
56	Today's recommended tutor: ben	03/25/2025	Looking for help with Trigonometry? ben@mail.com has you covered!	ADMIN
57	Today's recommended tutor: ben	03/25/2025	Looking for help with Trigonometry? ben@mail.com has you covered!	ADMIN

```
4 record(s) selected.
```

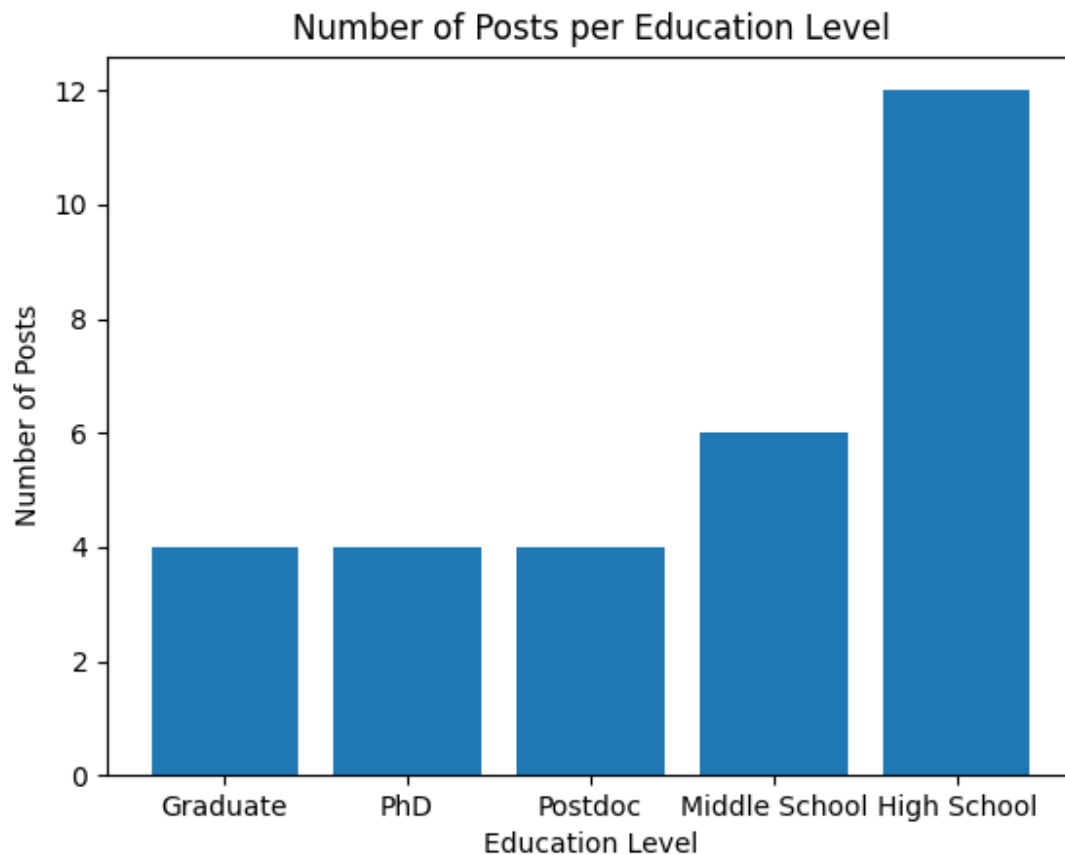
As in the above screenshot, the stored procedure was called with 'Trigonometry' as its argument, and it generated three posts recommending tutors that offer Trigonometry services.

Visualization

Vis 1

```
CONNECT TO DATABASE;
```

```
EXPORT TO DataVisualization/PostsPerEduLvl.csv OF DEL MODIFIED BY NOCHARDEL SELECT  
s.edu_level, COUNT(p.pID) AS post_count  
FROM Service s  
JOIN Advertises a ON s.email = a.email AND s.name = a.name  
JOIN Post p ON a.pID = p.pID  
GROUP BY s.edu_level  
ORDER BY COUNT(p.pID);
```

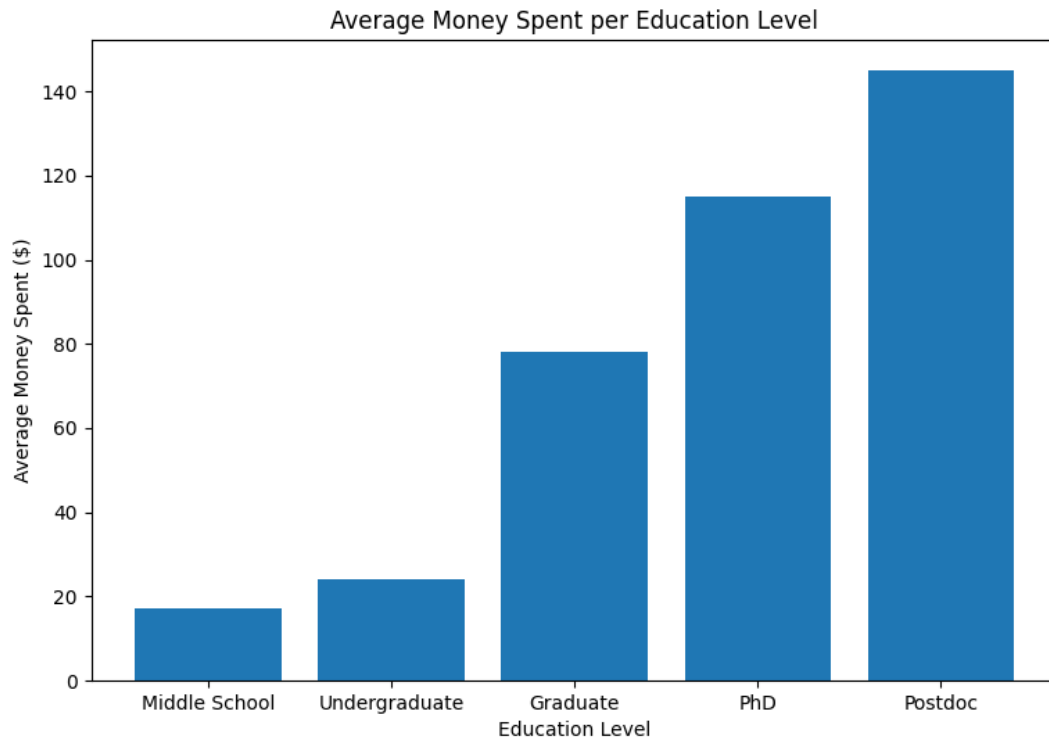


The above graph shows the number of posts per education level on our platform. As it is shown in the graph, the two education levels with the highest amounts of posts are Middle School and High School. This may be explained by the fact that they are the education levels with the most amount of students, hence more questions. Furthermore, High School may have more posts (compared to Middle School) as the material becomes more complicated and some parents may have a hard time helping their children.

Vis 2

```
CONNECT TO DATABASE;
```

```
EXPORT TO DataVisualization/SpentMoneyPerEduLvl.csv OF DEL MODIFIED BY NOCHARDEL
SELECT s.edu_level, AVG(s.price) AS avg_spent
FROM Service s
JOIN Purchase p ON s.email = p.email_seller AND s.name = p.name
GROUP BY s.edu_level
ORDER BY AVG(s.price);
```



The above graph shows the average amount of money spent on a service per education level. The graph shows that as the education level increases, the amount of money paid for a service increases as well. This may be true because as the education level increases, there are less resources online, hence the supply of Tutors becomes low and consequently the price of their help becomes high.

Python Script to Plot the Data

```
import matplotlib.pyplot as plt
import pandas as pd

# plot data from PostPerEduLvl
data = pd.read_csv("./PostsPerEduLvl.csv")

plt.bar(data.iloc[:, 0], data.iloc[:, 1])
plt.xlabel("Education Level")
plt.ylabel("Number of Posts")
plt.title("Number of Posts per Education Level")
```

```
plt.savefig("PostsPerEduLvl.png")

# plot data from SpentMoneyPerEduLvl
data = pd.read_csv("./SpentMoneyPerEduLvl.csv")

plt.figure(figsize=(9, 6))
plt.bar(data.iloc[:, 0], data.iloc[:, 1])
plt.xlabel("Education Level")
plt.ylabel("Average Money Spent ($)")
plt.title("Average Money Spent per Education Level")

plt.savefig("SpentMoneyPerEduLvl.png")
```

Creativity

For the creativity section, we decided to create a second stored process. This stored process takes as input the email of a student and enforces that they only purchase services of the same education level. The process checks the student's education level against each of their purchases and then deletes any discrepancies after adding them to the new relation cancelled_purchase with the following relational schema:

cancelled_purchase(email_buyer, email_seller, name, purchase_date, cancelled_date, price)
email_buyer references user
email_seller references user
name references service

The code can be found in the file creativity_procedure.sql

Queries in the database before running the process:

```
db2 => select * from student where email='david1@mail.com'
```

EMAIL	PAYMENT_INFO	EDU_LEVEL
david1@mail.com	-	High School

1 record(s) selected.

```
db2 => select s.email, s.name, s.subject, s.price, s.edu_level, p.purchase_date from service s join purchase p on s.email = p.email_seller where p.email_buyer = 'david1@mail.com' and p.name = s.name
```

EMAIL	PURCHASE_DATE	NAME	SUBJECT	PRICE	EDU_LEVEL
max@mail.com	06/23/2025	Tuesday Chem Tutorials	Chemistry Of Solutions	24	Undergraduate
ben@mail.com	03/11/2025	Functions notes p1	Functions	13	High School
ben@mail.com	01/05/2025	Basic Algebra	Mathematics	15	Middle School
max@mail.com	03/15/2025	Doctoral Chemistry	Chemistry	110	PHD

4 record(s) selected.

After running the process with example input 'david1@mail.com':

```
db2 => call enforce_level('david1@mail.com')
```

Return Status = 0

```
db2 => select s.email, s.name, s.subject, s.price, s.edu_level, p.purchase_date from service s join purchase p on s.email = p.email_seller where p.email_buyer = 'david1@mail.com' and p.name = s.name
```

EMAIL	PURCHASE_DATE	NAME	SUBJECT	PRICE	EDU_LEVEL
ben@mail.com	03/11/2025	Functions notes p1	Functions	13	High School

1 record(s) selected.

```
db2 => select * from cancelled_purchase
```

EMAIL_BUYER	EMAIL_SELLER	NAME	PURCHASE_DATE	CANCELLED_DATE	PRICE
david1@mail.com	max@mail.com	Tuesday Chem Tutorials	06/23/2025	03/25/2025	24
david1@mail.com	ben@mail.com	Basic Algebra	01/05/2025	03/25/2025	15
david1@mail.com	max@mail.com	Doctoral Chemistry	03/15/2025	03/25/2025	110

3 record(s) selected.

It can be seen that all of the services that are not of 'High School' level were deleted and corresponding entries were created in the cancelled_purchase table